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Conclusion

The primary goal of this analysis was to develop a stable, interpretable model that maximizes sensitivity to ensure that no patient with cancer is missed, a non-negotiable requirement for patient safety. We successfully leveraged a full suite of unsupervised learning and rule-mining techniques to engineer features that perfectly isolate the high-risk patient profile.

Logit Model:

The Logistic Regression model, using the same robust features (Cluster, MAD, and Rule Match flags), provided a stable, highly interpretable baseline but demonstrated clear limitations in maximizing the detection rate. While the model correctly flagged 71% of patients who actually had cancer, it couldn't overcome the extreme class imbalance and the non-linear way the high-risk patients are situated in the data.

KNN Model:

The K-Nearest Neighbors (KNN) model, using the exact same set of high-confidence binary features you engineered, proved to be the absolute winner by successfully maximizing sensitivity.

However,t he cost of this perfect detection rate was an increase in False Positives, specifically, an additional 6-ish patients (from 18 in the Logit to 24 in the KNN) were incorrectly flagged as high-risk.

Final Reccomendation

Implement the KNN model based on the three engineered binary features (K-Means Cluster, MAD Outlier, and Association Rule Match).

Justification:

1. Safety First: We hit 100% sensitivity. This means we meet the highest standard of patient safety by guaranteeing that every confirmed cancer case in the dataset is flagged for follow-up testing.
2. Acceptable Trade-Off: The cost of achieving this detection is flagging just 24 healthy patients (False Positives) for unnecessary follow-up. This is a small, justifiable operational cost compared to the catastrophic risk of missing a diagnosis.
3. Actionable Features: The model is not a black box. A patient is flagged because their profile matches one or more of three clear, unsupervised statistical signals, as they are either a multivariate outlier, a univariate outlier, or they perfectly match a high-confidence rule.

Thank you for your time.